

平衡的稳定条件

1. $\delta S = 0$ 平衡条件
 $\delta^2 S > 0$ 平衡的稳定条件

2. 从内能判据出发推导稳定条件.

$$\begin{cases} \delta U = 0 \\ \delta^2 U > 0 \\ \delta S = 0, \delta V = 0, \delta N = 0 \end{cases}$$

假设系统由两个系统组成.

$$U = U_1 + U_2 = \sum_{\alpha=1,2} U_\alpha$$

$$\delta U = \sum_{\alpha} \delta U_\alpha = \sum_{\alpha} (T_\alpha \delta S_\alpha - P_\alpha \delta V_\alpha + \mu_\alpha \delta N_\alpha)$$

取二阶微分.

$$\delta^2 U = \sum_{\alpha} [T_\alpha \delta^2 S_\alpha - P_\alpha \delta^2 V_\alpha + \mu_\alpha \delta^2 N_\alpha + \delta T_\alpha \delta S_\alpha - \delta P_\alpha \delta V_\alpha + \delta \mu_\alpha \delta N_\alpha]$$

$$\text{由 } S = S_1 + S_2 \quad V = V_1 + V_2 \quad N = N_1 + N_2$$

$$\delta S_2 = -\delta S_1 \quad \delta V_2 = -\delta V_1 \quad \delta N_2 = -\delta N_1$$

$$\delta^2 S_2 = -\delta^2 S_1 \quad \delta^2 V_2 = -\delta^2 V_1 \quad \delta^2 N_2 = -\delta^2 N_1$$

前三项:

$$(T_1 - T_2) \delta^2 S_1 - (P_1 - P_2) \delta^2 V_1 + (\mu_1 - \mu_2) \delta^2 N_1 = 0$$

$$\Rightarrow \delta^2 U = \sum_{\alpha} [\delta T_\alpha \delta S_\alpha - \delta P_\alpha \delta V_\alpha + \delta \mu_\alpha \delta N_\alpha]$$

$$S_{\alpha} = N_{\alpha} s_{\alpha} \quad V_{\alpha} = N_{\alpha} v_{\alpha}$$

s_{α}, v_{α} 为 1 mol 的熵和体积.

$$\delta S_{\alpha} = N_{\alpha} \delta s_{\alpha} + s_{\alpha} \delta N_{\alpha} \quad \delta V_{\alpha} = N_{\alpha} \delta v_{\alpha} + v_{\alpha} \delta N_{\alpha}$$

$$\Rightarrow \delta^2 U = \sum_{\alpha} [N_{\alpha} \delta s_{\alpha} \delta T_{\alpha} + s_{\alpha} \delta N_{\alpha} \delta T_{\alpha} - N_{\alpha} \delta v_{\alpha} \delta P_{\alpha} - v_{\alpha} \delta N_{\alpha} \delta P_{\alpha} + \delta U_{\alpha} \delta N_{\alpha}]$$

$$= \sum_{\alpha} N_{\alpha} (\delta s_{\alpha} \delta T_{\alpha} - \delta v_{\alpha} \delta P_{\alpha}) + \sum_{\alpha} \delta N_{\alpha} (s_{\alpha} \delta T_{\alpha} - v_{\alpha} \delta P_{\alpha} + \delta U_{\alpha})$$

$$\text{因为 } \delta U_{\alpha} = -s_{\alpha} \delta T_{\alpha} + v_{\alpha} \delta P_{\alpha}$$

$$\Rightarrow \delta^2 U = \sum_{\alpha} N_{\alpha} (\delta s_{\alpha} \delta T_{\alpha} - \delta v_{\alpha} \delta P_{\alpha}) > 0$$

$\delta^2 U$ 的正负应对任意的广延量 N_{α} 保持不变.

$$\Rightarrow \delta s_{\alpha} \delta T_{\alpha} - \delta v_{\alpha} \delta P_{\alpha} > 0 \quad \text{对任意 } \alpha \text{ 都成立.}$$

$$\text{省去下标} \quad \delta s \delta T - \delta v \delta P > 0$$

选 (v, T) 为独立变量.

$$\text{由 } \delta s = \left(\frac{\partial s}{\partial T}\right)_v \delta T + \left(\frac{\partial s}{\partial v}\right)_T \delta v$$

$$\delta p = \left(\frac{\partial p}{\partial T}\right)_v \delta T + \left(\frac{\partial p}{\partial v}\right)_T \delta v$$

$$\begin{aligned} & \left[\left(\frac{\partial s}{\partial T}\right)_v (\delta T)^2 + \left(\frac{\partial s}{\partial v}\right)_T \delta v \delta T \right] - \\ & \left[\left(\frac{\partial p}{\partial T}\right)_v \delta v \delta T + \left(\frac{\partial p}{\partial v}\right)_T (\delta v)^2 \right] > 0 \end{aligned}$$

利用麦氏关系 $(\frac{\partial S}{\partial V})_T = (\frac{\partial P}{\partial T})_V$

利用 $\begin{cases} C_V = \frac{dQ_V}{dT} = (\frac{\partial U}{\partial T})_V = T(\frac{\partial S}{\partial T})_V \\ \frac{C_V}{T} = (\frac{\partial S}{\partial T})_V \end{cases}$

$$\frac{C_V}{T} (\delta T)^2 - (\frac{\partial P}{\partial V})_T (\delta V)^2 > 0$$

$$\delta T, \delta V \text{ 独立} \Rightarrow C_V > 0, (\frac{\partial P}{\partial V})_T < 0$$

① $C_V = (\frac{\partial U}{\partial T})_V = \frac{dQ_V}{dT} > 0$ 表明, 如 1 区温度 T_1 高于 2 区温度 T_2 , 则 1 区的热量流向 2 区, T_1 下降, T_2 上升, 达到平衡条件 $T_1 = T_2$

② $(\frac{\partial P}{\partial V})_T < 0$ 表明:

如 V_1 增大, 则 P_1 减小, $P_1 < P_2$, 区域 1 会被压缩, 使 V_1 减小, P_1 增大, 达到 $P_1 = P_2$

如 $(\frac{\partial P}{\partial V})_T > 0$, V_1 增大使 P_1 增大, 则区域 1 会不断长大, 破坏平衡。

单元系的复相平衡

1. 单元系: 只有一种组元或一种分子。

① 设单元系有两个相 (α 和 β) 存在并达到平衡。

$$\text{平衡条件} \begin{cases} T^\alpha = T^\beta = T \\ p^\alpha = p^\beta = p \\ u^\alpha = u^\beta \end{cases}$$

选 T, p 为独立变量, 平衡条件写为

$$u^\alpha(T, p) = u^\beta(T, p)$$

在 $T-p$ 平面上是一条曲线

② 如三相同时存在并达到平衡, 则有

$$u^\alpha(T, p) = u^\beta(T, p) = u^\gamma(T, p)$$

曲线的交点称为三相点

2. 克拉珀龙方程

$T-p$ 平面上 $u^\alpha(T, p) = u^\beta(T, p)$ 曲线的确定主要靠实验测量.

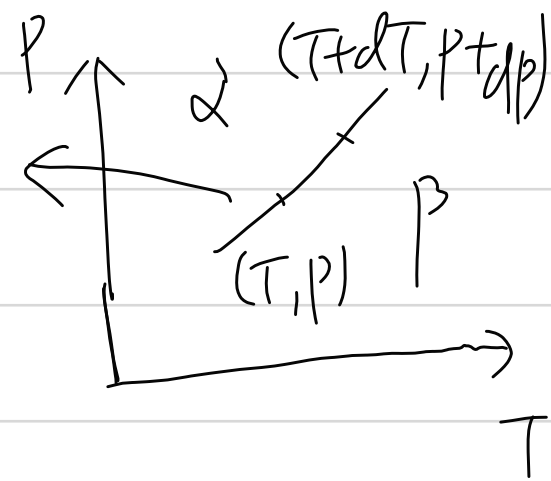
$$u^\alpha(T, p) = u^\beta(T, p)$$

$$p = p(T)$$

$$u^\alpha(T+dT, p+dp) = u^\beta(T+dT, p+dp)$$

$$du^\alpha = du^\beta$$

$$\begin{cases} du^\alpha = -s^\alpha dT + v^\alpha dp \\ du^\beta = -s^\beta dT + v^\beta dp \end{cases} \Rightarrow \frac{dp}{dT} = \frac{s^\alpha - s^\beta}{v^\alpha - v^\beta}$$



相变潜热 $\lambda_{\alpha\beta}$: 1 mol 物质在保持两相平衡的 T, p 不变的条件下, 从 β 相变为 α 相所吸收的热量.

$$\lambda_{\alpha\beta} = T(s^{\alpha} - s^{\beta})$$

$$\frac{dp}{dT} = \frac{\lambda_{\alpha\beta}}{T(v^{\alpha} - v^{\beta})}$$

有潜热和体积变化的相变 (一级相变)。